

What is statistical power and why is it important?

In this video, you will learn

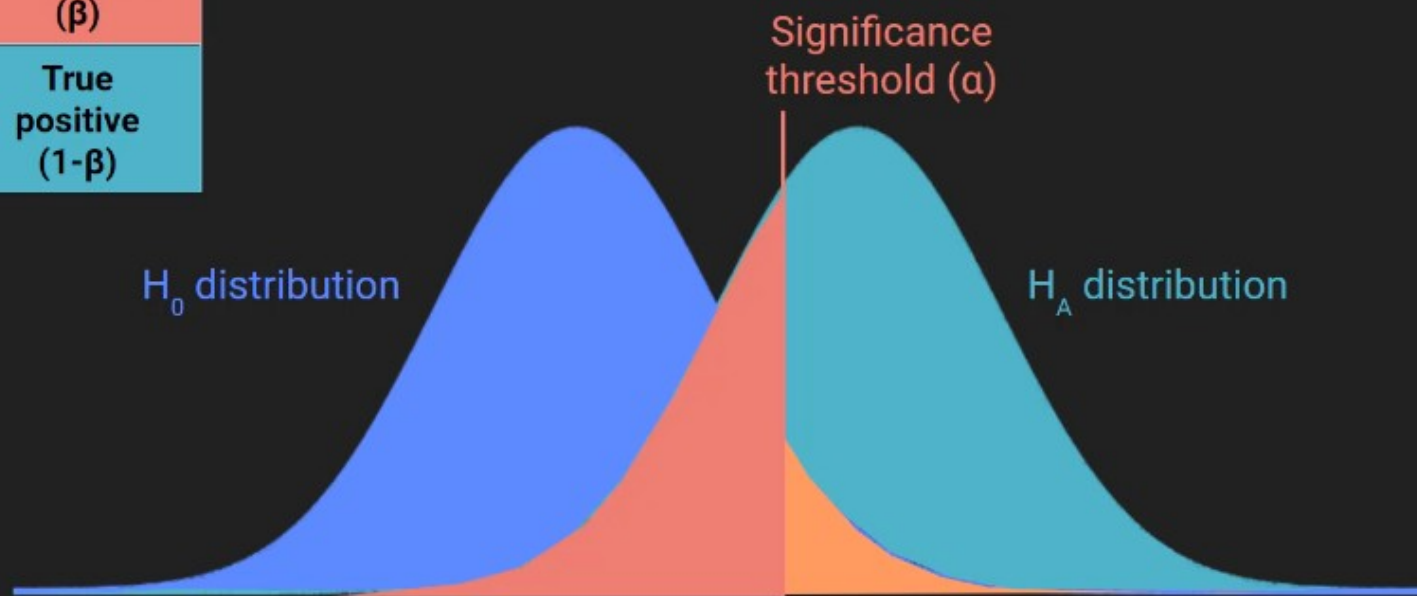
- What “statistical power” means and how it relates to Type II error probability.
- Cut-off ranges of power commonly used in psychology and behavioral sciences.
- The difficulties of controlling, calculating, and trusting statistical power calculations.

Statistical decisions

Reality about H_0

H_0 is true H_0 is false

Your decision	H_0 is true	H_0 is false
Don't reject H_0	True negative ($1-\alpha$)	Type II (false neg.) (β)
Reject H_0	Type I (false pos.) (α)	True positive ($1-\beta$)



Statistical decisions

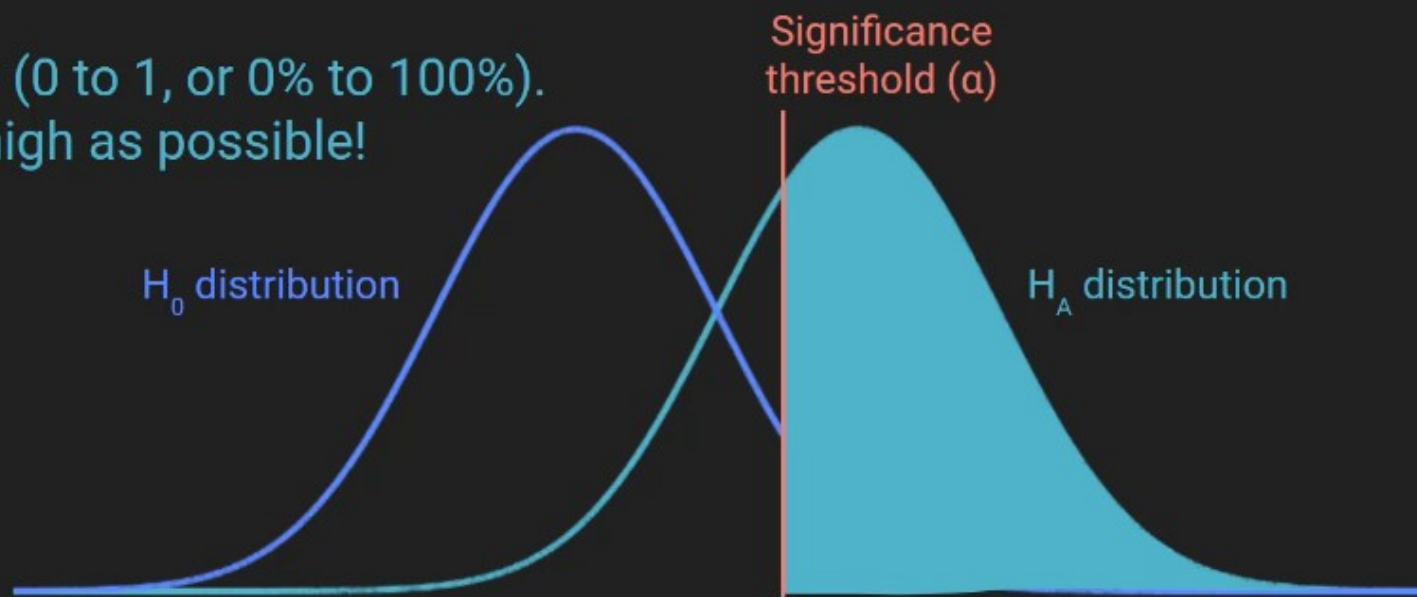
But what is statistical power really?

Power is the $p(\text{reject } H_0 \mid H_0 \text{ is false})$

a.k.a., the probability of finding an effect when it is really there.

Expressed as a probability (0 to 1, or 0% to 100%).

You want power to be as high as possible!



How to maximize power (statistical power)

Power increases with:

1. Sample size
2. Effect size
3. Lower α (e.g., $p < .1$)

Power decreases with:

1. Variability
2. Higher α (e.g., $p < .01$)

The problems with power

- Decreasing alpha increases power but also increases $p(\text{Type I})$.
- Some factors that influence power you can control; others you cannot (sample size, effect size, variability).
- Power can differ for different analyses in the same experiment.
- Computing “true power” requires knowing the true effect size.
- Power calculations from published studies are unreliable due to “publication bias” (non-significant findings are not reported).

Conclusion:

Statistical power is more of a useful guideline than a precise and trustworthy numerical value.

Estimating statistical power and sample size

In this video, you will learn

- How to compute statistical power for a basic t/z statistic.
- How to compute required sample size given a desired statistical power.
- The difference between “a priori power” and “post-hoc power.”

Formula for calculating power

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{(\bar{x} - \mu_0) \sqrt{n}}{\sigma}$$

z "Z" value for power (next slide)

$\bar{x}$ Effect size

μ_0 H_0 value

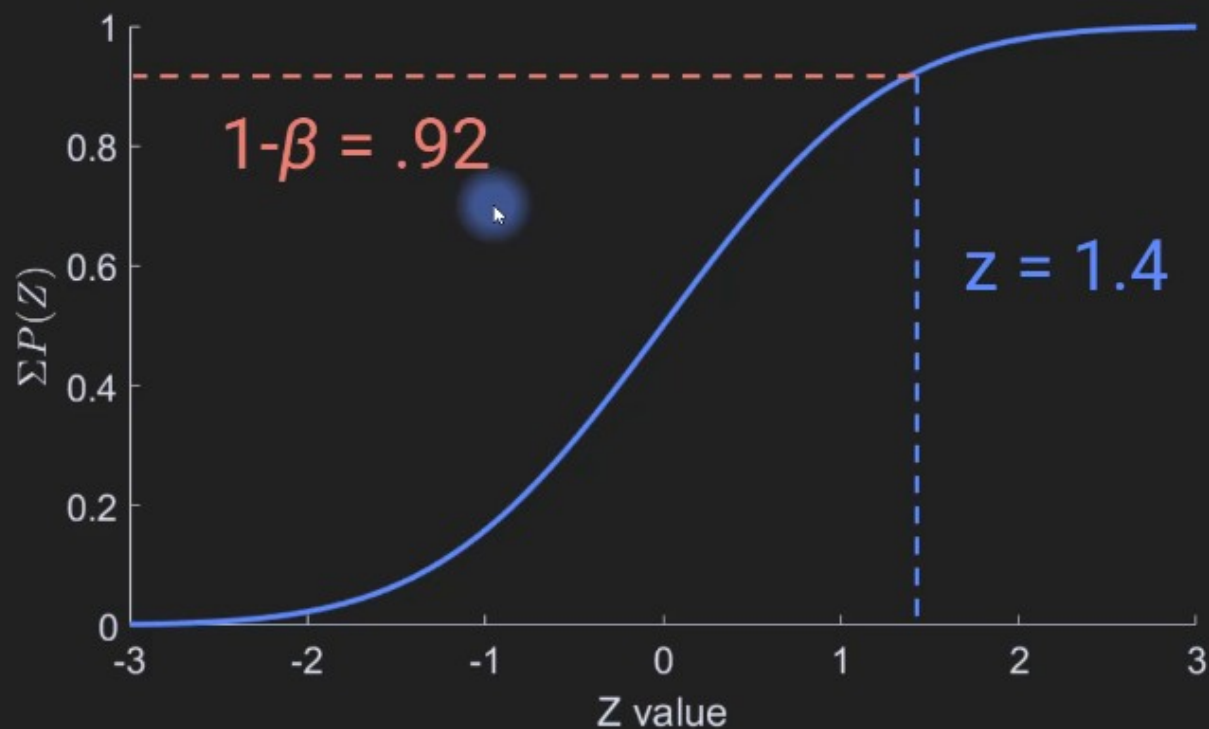
$\sigma / \sqrt{n}$ Standard error

Formula for calculating power

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{(\bar{x} - \mu_0) \sqrt{n}}{\sigma}$$

$$1 - \beta = 1 - p(Z > z)$$

Formula for calculating power



$$1 - \beta = 1 - p(Z > z)$$

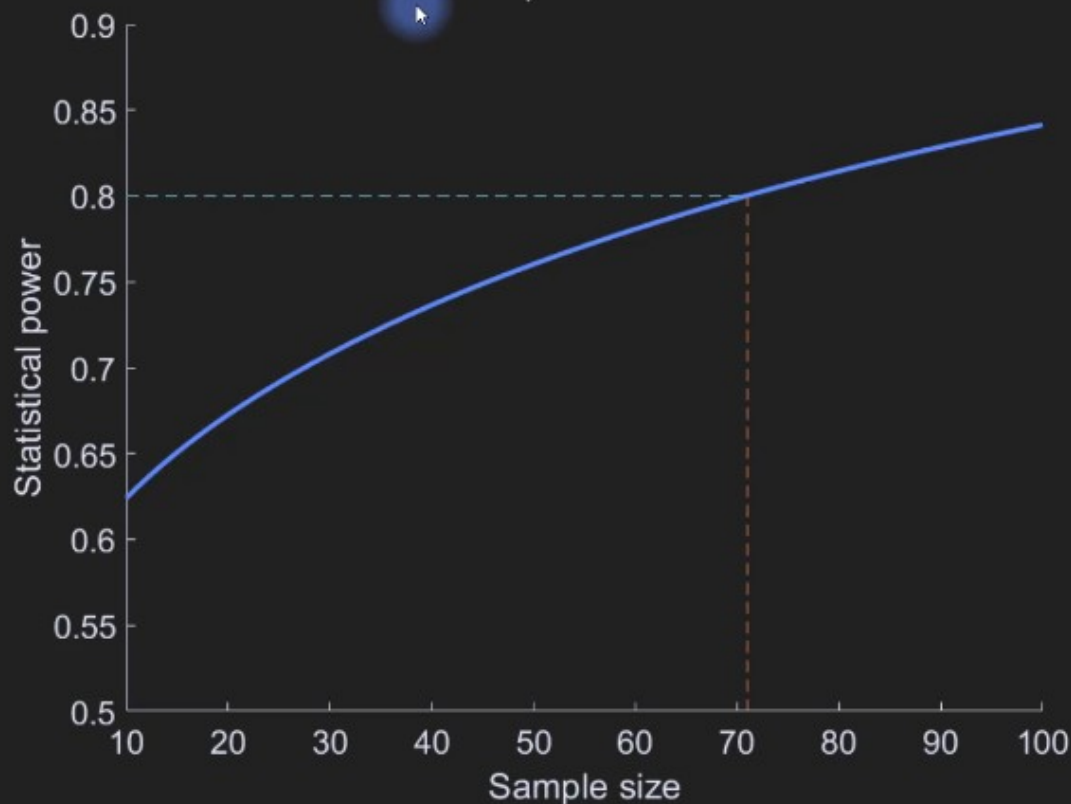
Formula for calculating sample size

$$z = \frac{(\bar{x} - \mu_0) \sqrt{n}}{\sigma}$$

$$n = \left(\frac{z \sigma}{\bar{x} - \mu_0} \right)^2$$

Example relationship between power and sample size

$$\bar{x} = 1, \quad \sigma = 10$$



Is it all so simple?

The formulas for more complicated statistical models (multiple regression, ANOVAs, etc.) are complicated, although the principal is the same.

In practice, best to use an online statistical power calculator.

A priori power vs. post-hoc power

A priori power

1. Get effect size and standard deviation from published studies or from pilot data.
2. Used to compute sample size prospectively.
3. Widely used and accepted, but wrought with difficulties and uncertainties.

Post-hoc power

1. Compute the power based on your completed study.
2. Proportional to the p-value, thus provides no new info.
3. Sometimes asked for by well-meaning people...



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